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How Important is Having Skin in the Game? Originator–Sponsor Affiliation and Losses on Mortgage-backed Securities

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1 Big picture

- **Question.** Does “skin in the game” improve the quality of securitized mortgages? More precisely, do mortgage-backed securities perform better when the loan originator is affiliated with the deal sponsor or retains servicing rights?
- **Setting.** The paper studies U.S. Alt-A residential mortgage-backed securities completed from **2003 to 2007**. Alt-A deals are a useful setting because they contain many low-documentation loans, where borrower quality depends more heavily on soft information and originator screening.
- **Core friction.** Securitization separates origination, sponsoring, servicing, rating, and investing. If originators sell loans and bear little future loss exposure, they may underinvest in screening.
- **Main measure of skin in the game.** The key proxy is **originator–sponsor affiliation**. If the originator is affiliated with the sponsor, the originating entity is more likely to retain exposure through the sponsor’s residual interest, junior tranches, or other first-loss positions. The paper also studies **originator–servicer affiliation**, because retaining servicing rights gives the originator an asset whose value falls when defaults rise.
- **Main empirical design.** Compare ex post performance, ex ante yields, and deal structure across affiliated, mixed, and unaffiliated MBS deals, controlling for observable mortgage risk, house price changes, and vintage fixed effects.
- **Baseline performance result.** Affiliated deals have much lower raw losses: mean cumulative net loss is **1.7%** for affiliated deals versus **4.7%** for unaffiliated deals and **4.9%** for mixed deals. In regressions, affiliated deals have about **0.62 percentage points** lower cumulative net loss, or about **18.8%** of the sample mean.
- **Where the effect is strongest.** The affiliation effect is concentrated in **low-documentation deals**. This is exactly where screening based on soft information should matter most.
- **Loan-level result.** At the loan level, affiliated loans default less often. In the raw data, default rates are **25.1%** for affiliated loans and **37.2%** for unaffiliated loans. In logit regressions, affiliation lowers default likelihood for low-documentation loans, even with detailed loan controls and originator fixed effects.
- **Identification strategy.** The paper uses several approaches: rich controls, propensity-score matching, instrumental variables based on sponsor past affiliation patterns, loan-level fixed effects, and a difference-in-differences design around acquisitions that changed originator affiliation status.
- **Pricing result.** Investors and rating agencies appear to recognize the incentive effect. Low-documentation affiliated deals have lower ex ante yields and require less credit enhancement. The average initial yield spread is about **32 basis points** lower for affiliated low-doc deals.
- **Economic takeaway.** Securitization problems are not only about whether a loan is securitized. They also depend on how much exposure the originator retains after securitization. Even within the securitized market, incentive alignment predicts both ex post loan performance and ex ante security pricing.

2 Data and measurement (institutional context and key objects)

2.1 Why Alt-A mortgage-backed securities are the right laboratory

- Alt-A mortgages sit between prime and subprime loans. Borrowers often have better credit scores than subprime borrowers, but the loans frequently fail to meet government-sponsored-enterprise underwriting guidelines.
- A defining feature of Alt-A pools is the high share of low- or no-documentation loans.
- Low-documentation lending is precisely where originators' private screening and soft information should be valuable.
- The paper therefore studies whether the same broad securitization market produces different outcomes depending on the originator's retained exposure.

Interpretation. The paper does not simply ask whether securitized loans perform worse than portfolio loans. It asks whether securitized loans perform differently depending on the contractual and organizational links among the originator, sponsor, and servicer.

2.2 Securitization players and the source of incentives

- **Originator.** Extends mortgage credit to borrowers and initially bears warehousing risk before the loan is sold.
- **Sponsor.** Purchases loans, structures the securitization, consults with rating agencies, files documents, and typically retains the most junior residual interest or subordinated position.
- **Servicer.** Collects payments, handles customer service, advances payments, and manages delinquency, workout, and foreclosure processes.
- **Investors.** Buy the tranches created from the mortgage pool.

Interpretation. Originator-sponsor affiliation matters because the sponsor's retained residual or junior position exposes the affiliated originator to losses. Originator-servicer affiliation matters because servicing rights are worth less when loans default or prepay quickly.

2.3 MBS sample construction

- The paper starts from ABSNet and identifies Alt-A deals using deal names and collateral descriptions.
- The initial universe contains **829** Alt-A deals completed from **2003 to 2007**.
- The authors remove second-lien securitizations, resecuritizations, deals with missing prospectuses, deals with missing originator identity, deals missing performance data, and deals missing key risk information.

- The final deal-level sample contains **474 Alt-A deals** backed by about **\$345 billion** in mortgages.
- Deal characteristics, originator identities, servicer identities, and structure variables are hand-collected from offering prospectuses.
- Deal performance and initial yields come from ABSNet. Loan-level data come from ABSNet Loan. House price changes come from FHFA state- and MSA-level house price indices.

Interpretation. The data construction is labor-intensive because affiliation is not a simple database field. The authors need prospectus-level information on who originated, sponsored, and serviced the loans.

2.4 Affiliation measures

- **Affiliated deal.** All loans in the securitization were originated by entities affiliated with the sponsor.
- **Unaffiliated deal.** All loans were originated by entities unaffiliated with the sponsor.
- **Mixed deal.** Some, but not all, loans were originated by entities affiliated with the sponsor.
- **Continuous sponsor-originated share.** The percentage of the mortgage balance originated by the sponsor or its affiliates.
- **Originator-servicer affiliation.** The percentage of the mortgage balance serviced by affiliates of the originators.
- **Sponsor-servicer affiliation.** The percentage of the mortgage balance serviced by the sponsor.

Interpretation. The discrete affiliated/unaffiliated/mixed classification captures the cleanest organizational distinction, while the continuous measures allow the paper to use variation in partial affiliation.

2.5 Summary facts on affiliation

- In the full sample, **48.7%** of deals are affiliated, **31.0%** are mixed, and **20.3%** are unaffiliated.
- The average percentage of mortgages originated by the sponsor is **66.7%**.
- The average percentage of mortgages serviced by the originators is **79.6%**.
- Affiliation changes over time: affiliated deals fall from around **60%** of deals early in the sample to around **40%** in 2007, while mixed deals become more common.

Interpretation. The data show meaningful variation in retained exposure. Many originators retained some skin in the game, but not always, and this variation is central to the tests.

2.6 Main performance variables

- **Cumulative net loss rate.** Losses of principal net of recoveries through August 2009, divided by original deal balance.
- **Foreclosure rate.** Face value of loans in foreclosure at the end of August 2009, divided by end-of-month pool balance.
- **Loan default.** At the loan level, a loan is classified as defaulted if it becomes **60+ days delinquent** between securitization and August 2009.

Interpretation. Cumulative net loss is the cleanest realized-loss measure. Foreclosure captures pipeline losses not yet fully realized. Loan-level default allows the authors to use detailed borrower and loan controls.

2.7 Main risk controls

- FICO score.
- Loan-to-value ratio.
- Share of adjustable-rate mortgages.
- Negative amortization indicator.
- Low-documentation indicator.
- Share of purchase loans.
- Share of family-owned properties.
- Share of primary-residence loans.
- House price changes between deal close or loan origination and August 2009.
- Vintage fixed effects.

Interpretation. Affiliated and unaffiliated deals differ along observable risk dimensions, so the regression design must separate incentive effects from differences in loan composition.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework

- If affiliation gives originators more exposure to future pool performance, affiliation should increase screening effort.

- The screening effect should be strongest for loans where soft information is useful, especially low-documentation loans.
- If investors and rating agencies understand this incentive channel, affiliated deals should receive lower required yields and require less credit enhancement.

Interpretation. The paper links three outcomes: screening incentives, realized performance, and ex ante pricing/structure. A convincing incentive story should show all three moving in the same direction.

3.2 Baseline deal-level performance regression

The core deal-level specification can be written as

$$Performance_d = \alpha + \beta Affiliated_d + X'_d \gamma + \lambda_{v(d)} + \varepsilon_d, \quad (1)$$

where $Performance_d$ is either cumulative net loss or foreclosure rate, $Affiliated_d$ is a dummy for full originator–sponsor affiliation, X_d includes deal risk controls and house price changes, and $\lambda_{v(d)}$ are vintage fixed effects.

Interpretation. The coefficient β compares affiliated deals with unaffiliated and mixed deals after holding fixed observable collateral risk and securitization vintage. A negative β means affiliated deals lose less money or have fewer foreclosures.

3.3 Alternative affiliation specifications

The authors also estimate versions using unaffiliated and mixed deal dummies:

$$Performance_d = \alpha + \beta_1 Unaffiliated_d + \beta_2 Mixed_d + X'_d \gamma + \lambda_{v(d)} + \varepsilon_d, \quad (2)$$

with affiliated deals as the omitted category.

They also use a continuous measure:

$$Performance_d = \alpha + \beta ShareOriginatedBySponsor_d + X'_d \gamma + \lambda_{v(d)} + \varepsilon_d. \quad (3)$$

Interpretation. These specifications test whether the result is tied to a particular binary definition. If skin in the game matters, losses should be higher for unaffiliated and mixed deals and lower when a larger share of the pool is originated by the sponsor.

3.4 Low-doc versus high-doc split

The baseline model is estimated separately for low-documentation and high-documentation deals:

$$Performance_d = \alpha_g + \beta_g Affiliated_d + X'_d \gamma_g + \lambda_{v(d),g} + \varepsilon_{d,g}, \quad g \in \{LowDoc, HighDoc\}. \quad (4)$$

Interpretation. The key prediction is $\beta_{LowDoc} < 0$ and economically large, while $\beta_{HighDoc}$ should be small or insignificant. This heterogeneity is important because it distinguishes a screening story from a generic sponsor-quality or deal-quality story.

3.5 Propensity-score matching

The paper estimates a logit model for affiliation:

$$Pr(Affiliated_d = 1) = \Lambda(X'_d\theta + \lambda_{v(d)}), \quad (5)$$

then matches each affiliated deal to an unaffiliated or mixed deal with a similar propensity score.

The treatment effect on affiliated deals is

$$ATT = E[Performance_d(1) - Performance_d(0) | Affiliated_d = 1]. \quad (6)$$

Interpretation. Matching asks whether affiliated deals perform better than observably similar unaffiliated/mixed deals, reducing concern that the regression result is driven by functional-form assumptions.

3.6 Instrumental-variables design

The IV approach instruments current affiliation using the sponsor's average affiliation rate in its past sample deals:

$$Affiliated_d = \pi_0 + \pi_1 PastAffiliationRate_{s(d), t < d} + X'_d\pi + u_d, \quad (7)$$

$$Performance_d = \alpha + \beta \widehat{Affiliated}_d + X'_d\gamma + \varepsilon_d. \quad (8)$$

Interpretation. The instrument captures persistent sponsor-level tendencies to create affiliated deals. The identifying assumption is that past sponsor affiliation patterns affect current deal performance mainly through current affiliation, not through unobserved current loan risk.

3.7 Loan-level default regressions

At the loan level, the paper estimates logit models of default:

$$Pr(Default_i = 1) = \Lambda(\alpha + \beta AffiliatedLoan_i + X'_i\gamma + \delta_o + \eta_{state/vintage}), \quad (9)$$

where X_i contains loan-level risk controls and some specifications include originator fixed effects δ_o .

Interpretation. Loan-level regressions help with measurement. Instead of using deal averages, the authors compare individual loans with similar FICO, LTV, mortgage type, occupancy, house price changes, and other risk characteristics.

3.8 Difference-in-differences around acquisitions

Two acquisitions changed affiliation status during the sample: MortgageIT was acquired by Deutsche Bank and First Franklin was acquired by Merrill Lynch. The paper estimates

$$Pr(Default_i = 1) = \Lambda(\alpha + \beta_1 AcquiringSponsor_i + \beta_2 PostAcquisition_i + \beta_3 AcquiringSponsor_i \times PostAcquisition_i + X_i' \theta). \quad (10)$$

Interpretation. The interaction term compares loans sold to the future/acquiring sponsor before versus after the acquisition, relative to loans sold to other unaffiliated sponsors. A negative interaction means loans perform better once the originator becomes affiliated with the sponsor.

3.9 Originator-servicer affiliation regressions

For deals in which originator and sponsor are not affiliated, the paper estimates

$$Performance_d = \alpha + \beta ShareServicedByOriginator_d + X_d' \gamma + \lambda_{v(d)} + \varepsilon_d. \quad (11)$$

It also tests sponsor-servicer affiliation.

Interpretation. This isolates the servicing-rights channel from the sponsor-affiliation channel. If retaining servicing rights provides skin in the game, higher originator-servicer affiliation should predict lower losses, especially in low-doc deals.

3.10 Pricing and deal-structure regressions

The ex ante pricing and structure tests use specifications of the form

$$DealStructure_d = \alpha + \beta Affiliated_d + X_d' \gamma + \lambda_{v(d)} + \varepsilon_d, \quad (12)$$

where $DealStructure_d$ is initial average yield spread, subordination, overcollateralization probability, or overcollateralization target.

For overcollateralization probability, the paper uses a logit model:

$$Pr(OC_d = 1) = \Lambda(\alpha + \beta Affiliated_d + X_d' \gamma + \lambda_{v(d)}). \quad (13)$$

Interpretation. If investors and rating agencies expect affiliated deals to perform better, affiliated deals should have lower yield spreads and need less protection for senior securities.

3.11 AAA-bond yield spread regression

To rule out the possibility that deal-level yield differences only reflect differences in ratings composition, the paper studies fixed-rate AAA tranches:

$$Spread_b = \alpha + \beta Affiliated_{d(b)} + \delta WAL_b + \psi WAC_{d(b)} + \lambda_{v(b)} + \tau_{type(b)} + \varepsilon_b. \quad (14)$$

Interpretation. This compares yields within the AAA category. If AAA bonds from affiliated low-doc deals still have lower spreads, investors price incentive differences even within the same rating class.

4 Identification and empirical design (what makes the estimates credible)

4.1 Main identifying comparison

- The central comparison is between affiliated and unaffiliated/mixed securitizations with similar observable collateral characteristics and the same broad vintage environment.
- The cleanest identifying variation is strongest where theory predicts it should be: low-documentation mortgages.
- The pricing tests give an ex ante counterpart: investors and rating agencies also seem to distinguish affiliated from unaffiliated deals before outcomes are realized.

Interpretation. The paper's argument is not based on one coefficient alone. It is based on a pattern across performance, low-doc heterogeneity, loan-level tests, acquisition shocks, and ex ante pricing.

4.2 Observable risk selection

- Unaffiliated and mixed deals are riskier on several observable dimensions: lower FICO, higher LTV, more adjustable-rate loans, more negative amortization, and more low-documentation loans.
- Baseline regressions control for these variables directly.
- Propensity-score matching compares affiliated deals with unaffiliated/mixed deals that have similar observable risk profiles.

Interpretation. Observable selection is real, but the paper shows that the affiliation effect remains after comparing more similar deals and loans.

4.3 Unobservable selection and endogeneity

- Affiliation is a choice, so unobserved loan quality could still differ across affiliated and unaffiliated deals.
- The IV design uses sponsor past affiliation patterns as an instrument for current affiliation.
- The IV estimates are more negative than OLS, suggesting that simple regressions may understate the effect of affiliation rather than exaggerate it.

- Loan-level specifications with originator fixed effects compare affiliated and unaffiliated loans originated by the same institution.

Interpretation. These tests reduce the concern that the results only reflect persistent differences in sponsor quality or originator underwriting standards.

4.4 Natural experiment logic

- The acquisitions of MortgageIT by Deutsche Bank and First Franklin by Merrill Lynch changed whether loans sold to the acquiring sponsor were affiliated.
- The difference-in-differences design compares loans sold to acquiring sponsors before and after acquisition against loans sold to other sponsors.
- The interaction effect is significant for low-documentation loans but not full-documentation loans.

Interpretation. The acquisition evidence is especially useful because it uses a change in organizational affiliation rather than only cross-sectional differences across deals.

4.5 Why low-documentation heterogeneity is central

- Low-doc loans rely less on hard verified income and asset information.
- Screening based on private or soft information should therefore matter more.
- The paper consistently finds stronger affiliation effects in low-doc deals and low-doc loans.

Interpretation. This cross-sectional pattern is hard to reconcile with a pure story that affiliated sponsors are simply better in all dimensions. It fits the screening-incentive mechanism.

4.6 Pricing and structure as validation

- Affiliated low-doc deals have lower initial yield spreads and require less subordination.
- They are also less likely to require overcollateralization.
- Even fixed-rate AAA tranches from affiliated low-doc deals have lower yield spreads.

Interpretation. Investors and rating agencies appear to price affiliation before the realized losses occur. This supports the idea that affiliation contains economically meaningful information about incentives.

4.7 Main limitations

- Affiliation is an indirect proxy for retained exposure. The paper does not observe every position held by the originator, sponsor, or affiliated entities over the full life of the security.

- The main deal-level sample focuses on Alt-A deals, though the paper also checks broader ABSNet Loan samples of subprime and Alt-A loans.
- IV validity depends on the assumption that sponsor past affiliation patterns do not directly predict unobserved current deal performance except through current affiliation.
- The paper shows relative pricing within the MBS market, not that the whole market correctly priced aggregate housing risk.

Interpretation. The evidence supports the importance of incentive alignment inside securitization, but it does not claim that MBS markets were globally efficient or that all mortgage risk was correctly priced before the crisis.

5 Tables and results

5.1 Table 1: Examples of affiliated, unaffiliated, and mixed deals

Read:

- Table 1 gives concrete examples of how deals are classified.
- Affiliated examples include deals in which Bank of America, Countrywide, Lehman Brothers, or Washington Mutual originated **100%** of the loans and were affiliated with the sponsor.
- Unaffiliated examples include deals where the sponsor bought loans from outside originators such as GreenPoint, National City, or IndyMac.
- Mixed examples combine affiliated and unaffiliated originators inside the same trust.

Economic message. Affiliation is not a theoretical label. It is observable in deal documents and varies meaningfully across securitizations.

5.2 Table 2: Summary statistics for affiliation measures

Read:

- Out of **474** deals, **48.7%** are affiliated, **31.0%** are mixed, and **20.3%** are unaffiliated.
- The average share of mortgages originated by the sponsor is **66.7%**.
- The average share of mortgages serviced by originators is **79.6%**.
- The affiliated share declines over time, while mixed deals rise, making vintage controls important.

Economic message. The market contains enough variation in organizational form to test whether retained exposure matters for performance.

Table 1
Examples of affiliated, unaffiliated, and mixed deals in our sample

Deal name	Deal type	Originator(s)	Originator affiliated with sponsor?	Percent originated (by \$ amount) by originator	Percent serviced (by \$ amount) by originator	Percent serviced (by \$ amount) by sponsor	
Banc of America Alternative Loan Trust 2007-1	Affiliated	Bank of America	Yes	100.0%	100.0%	100.0%	
Countrywide ALT 2006-HY3		Countrywide	Yes	100.0%	100.0%	100.0%	
Lehman XS Trust 2007-178		Lehman Brothers	Yes	100.0%	100.0%	100.0%	
WMAI Series 2007-4		Washington Mutual	Yes	100.0%	100.0%	100.0%	
New Century Alternative Mortgage Loan Trust 2006-ALT1		New Century	Yes	100.0%	0.0%	0.0%	
Bear Stearns Alt-A Trust 2005-1		Greentop	No	100.0%	100.0%	0.0%	
Countrywide ALT 2007-ALT-1		Unaffiliated	No	100.0%	0.0%	0.0%	
Deutsche Alt-A Securities Mortgage Loan Trust 2003-1		Unaffiliated	Alliance Bancorp	No	80.0%	80.0%	0.0%
Merrill Lynch Alternative Loan Trust 2007-OAR2		Unaffiliated	National City Mortgage	No	19.6%	19.6%	0.0%
			Chase Manhattan	No	0.4%	0.4%	0.0%
			IndyMac	Yes	51.9%	51.9%	6.3%
			Countrywide	No	41.8%	41.8%	0.0%
			Bayrock Mortgage Corp	No	6.3%	0.0%	0.0%
			Washington Mutual	Yes	12.7%	100.0%	100.0%
			Greentop	No	28.1%	0.0%	0.0%
			American Home Mortgage	No	35.7%	0.0%	0.0%
			Other (-10%)	No	12.5%	0.0%	0.0%
			Countrywide	Yes	67.3%	100.0%	100.0%
Countrywide ALT 2006-OA18		Mixed	MortgageIT	No	25.1%	0.0%	0.0%
			Other (-10%)	No	7.6%	0.0%	0.0%
J.P. Morgan Alternative Loan Trust 2006-51		Mixed	J.P. Morgan	Yes	57.0%	54.7%	54.7%
			SunTrust	No	21.4%	21.4%	0.0%
			Greentop	No	17.7%	0.0%	0.0%
Lehman XS Trust 2007-208		Mixed	Countrywide	No	15.9%	15.9%	0.0%
			Other (-10%)	No	8.0%	8.0%	0.0%
			Lehman Brothers	Yes	46.7%	99.6%	99.6%
			Quicken Loans	No	27.6%	0.0%	0.0%
			American Home Mortgage	No	24.3%	0.0%	0.0%
			Greentop	No	0.9%	0.0%	0.0%
			Countrywide	No	0.4%	0.4%	0.0%

5.3 Table 3: Univariate deal characteristics and performance

Read:

- Affiliated deals have higher average FICO scores, lower LTVs, and fewer risky features.
- Mean cumulative net loss is **1.7%** for affiliated deals, **4.7%** for unaffiliated deals, and **4.9%** for mixed deals.
- Mean foreclosure rate is **8.4%** for affiliated deals, **16.3%** for unaffiliated deals, and **14.2%** for mixed deals.

Economic message. Affiliated deals perform better in raw data, but affiliated deals are also observably different. This motivates the multivariate analysis.

Table 3
Univariate deal-level summary statistics and determinants of originator-sponsor affiliation

Variables	All (N=474)			Affiliated (N=231)			Unaffiliated (N=96)			Mixed (N=147)		
	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD
Vintages:												
2003 (d)	8.4%	—	—	10.4%	—	—	9.4%	—	—	4.8%	—	—
2004 (d)	7.8%	—	—	10.4%	—	—	5.2%	—	—	5.4%	—	—
2006 (d)	14.6%	—	—	13.4%	—	—	13.5%	—	—	17.0%	—	—
2008 (d)	41.4%	—	—	42.9%	—	—	40.6%	—	—	39.5%	—	—
2007 (d)	27.8%	—	—	22.9%	—	—	31.3%	—	—	31.3%	—	—
Mortgage characteristics:												
FICO	710.9	710.5	12.8	714.3	714.0	12.5	705.1	706.0	12.5	709.4	709.2	12.1
LTV	71.5	71.6	4.8	71.1	71.5	4.0	75.9	75.4	3.2	75.6	75.7	5.1
% adjustable rate	42.7	0.0	41.2	18.5	0.0	38.9	60.9	100.0	46.5	68.9	100.0	40.2
100 * Negative amortization (d)	15.0	0.0	35.7	7.4	0.0	26.2	34.4	0.0	47.7	14.3	0.0	35.1
% low documentation	76.3	77.3	12.0	72.7	72.6	10.8	82.7	86.0	13.2	77.6	86.4	10.9
% purchase loans	49.9	46.1	16.0	41.9	41.5	13.2	43.5	46.4	16.6	53.8	58.1	16.7
% family owned	62.7	62.1	7.2	63.8	62.9	6.7	61.8	61.6	6.4	61.5	59.9	8.3
% primary residence	77.7	82.3	14.1	77.0	84.3	17.4	79.1	81.2	11.1	78.0	80.9	9.6
Performance and house prices:												
Cumulative net loss rate (%)	3.3	2.2	3.3	1.7	1.3	1.8	4.7	4.6	3.4	4.9	3.8	3.8
Foreclosure rate (%)	11.8	11.5	7.3	8.4	7.8	5.4	16.3	16.8	8.4	14.2	14.8	6.7
Pre-deal house price run-up (%)	7.4	7.7	4.4	7.8	8.1	4.1	7.4	7.6	4.7	6.7	7.2	4.6
House price change (%)	-4.0	-8.9	14.4	-1.8	-6.7	14.9	-5.6	-10.3	15.4	-6.3	-10.1	12.3

This table presents deal-level summary statistics for securitized mortgages backing our sample deals. We present summary statistics for the overall sample and by originator-sponsor affiliation. Variable definitions are in Appendix 2.

Table 2
Summary statistics for affiliation measures

Panel A: Summary statistics for all sample periods

Variables	N	Mean	SD	Min	Median	Max
All deals:						
Affiliated (d)	474	48.7%	—	—	—	—
Mixed (d)	474	31.0%	—	—	—	—
Unaffiliated (d)	474	20.3%	—	—	—	—
% of mortgages originated by the sponsor	474	66.7%	41.4%	0.0%	97.0%	100.0%
% of mortgages serviced by the originator(s)	471	79.6%	37.5%	0.0%	100.0%	100.0%
% of mortgages serviced by the sponsor	471	78.0%	38.2%	0.0%	100.0%	100.0%
Number of originators	474	2.6	2.2	1.0	1.0	11.0
Affiliated deals:						
% of mortgages serviced by the originator(s)	231	87.9%	32.7%	0.0%	100.0%	100.0%
% of mortgages serviced by the sponsor	231	99.1%	9.3%	0.0%	100.0%	100.0%
Unaffiliated deals:						
% of mortgages serviced by the originator(s)	96	38.7%	40.8%	0.0%	26.8%	100.0%
% of mortgages serviced by the sponsor	96	30.3%	44.1%	0.0%	0.0%	100.0%
Number of originators	96	4.8	2.7	1.0	4.0	11.0
Mixed deals:						
% originated by the sponsor	147	58.0%	28.0%	1.0%	65.4%	98.9%
% of mortgages serviced by the originator(s)	144	93.5%	19.2%	0.0%	100.0%	100.0%
% of mortgages serviced by the sponsor	144	76.0%	33.4%	0.0%	100.0%	100.0%
Number of originators	147	3.6	1.9	1.0	3.0	11.0

Panel B: Mean characteristics by vintage

Variables	Vintage				
	2003	2004	2005	2006	2007
Affiliated (d)	60.0%	64.9%	44.9%	50.5%	40.2%
Mixed (d)	17.5%	21.6%	36.2%	29.6%	37.1%
Unaffiliated (d)	22.5%	13.5%	18.8%	19.9%	22.7%
% originated by the sponsor	74.3%	75.5%	70.2%	64.5%	63.5%
% of mortgages serviced by the originator(s)	82.6%	88.1%	87.8%	76.6%	76.6%
% of mortgages serviced by the sponsor	75.2%	78.4%	74.4%	81.8%	75.0%
# of originators	2.0	2.2	2.3	2.8	2.6
Number of deals	40	37	69	196	132

This table presents summary statistics for various originator affiliation measures for the 474 Alt-A deals in our sample. We use (d) to denote that the variable is a dummy variable. In Panel A, we present the summary statistics for the overall sample. In Panel B, we present mean values of the affiliation measures by vintage. We use (d) to denote that the variable is a dummy variable. Variable definitions are in Appendix 2.

Table 4
Originator-sponsor affiliation and deal performance

Explanatory variables	Cumulative net loss rate (%)					Foreclosure rate (%)				
	All (1)	All (2)	All (4)	Low-doc (5)	High-doc (6)	All (7)	All (8)	All (9)	High-doc (10)	High-doc (11)
Affiliated (d)	-0.42*** (-3.11)	—	—	-1.510*** (-4.41)	0.12 (0.51)	-1.37*** (-3.45)	—	—	-2.83*** (-3.80)	-4.20 (-0.38)
Unaffiliated (d)	—	0.06** (2.32)	—	—	—	—	3.11*** (5.53)	—	—	—
Mixed (d)	—	0.56** (2.55)	—	—	—	—	0.07 (0.16)	—	—	—
% originated by the sponsor	—	—	-0.55** (-2.02)	—	—	—	—	-3.40*** (-6.58)	—	—
Low documentation (d)	0.55*** (2.72)	0.53*** (2.64)	0.62*** (3.07)	—	—	1.38*** (7.58)	1.02** (-1.79)	1.20*** (2.38)	—	—
FICO	-0.05*** (-1.44)	-0.05*** (-1.42)	-0.05*** (-1.48)	-0.03** (-2.36)	-0.04** (-2.57)	-0.15*** (-3.45)	-0.15*** (-3.45)	-0.15*** (-3.45)	-0.16*** (-3.45)	-0.14*** (-3.45)
LTV	0.02*** (3.23)	0.02*** (3.18)	0.02*** (3.18)	0.01 (0.67)	0.02*** (3.86)	0.05*** (4.02)	0.05*** (2.88)	0.05*** (2.15)	0.05*** (1.76)	0.05*** (1.40)
% adjustable rate	-1.09** (8.02)	-1.09** (8.07)	-1.09** (8.51)	-0.73 (0.51)	-0.73 (0.51)	-3.47*** (9.07)	-3.47*** (9.07)	-3.47*** (9.07)	-3.47*** (9.07)	-3.47*** (9.07)
Negative amortization (d)	-2.00 (1.81)	-2.01 (1.79)	-2.01 (1.81)	-0.94 (2.32)	-0.94 (2.32)	-3.47*** (9.07)	-3.47*** (9.07)	-3.47*** (9.07)	-3.47*** (9.07)	-3.47*** (9.07)
% purchase loans	0.01 (0.62)	0.01 (0.62)	0.01 (0.62)	0.04 (1.05)	0.04 (1.05)	0.05*** (2.30)	0.05*** (2.30)	0.05*** (2.30)	0.05*** (2.30)	0.05*** (2.30)
% family owned	-0.04*** (-3.50)	-0.04*** (-3.51)	-0.04*** (-3.51)	-0.04*** (-3.83)	-0.04*** (-3.83)	-0.02 (-1.59)	-0.02 (-1.59)	-0.02 (-1.59)	-0.02 (-1.59)	-0.02 (-1.59)
House price change	-0.10*** (-4.22)	-0.10*** (-4.20)	-0.10*** (-4.06)	-0.22*** (-4.42)	-0.22*** (-4.42)	-0.24*** (-5.03)	-0.24*** (-5.03)	-0.24*** (-5.03)	-0.24*** (-5.03)	-0.24*** (-5.03)
Vintage dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	474	474	474	237	237	474	474	474	237	237
Adjusted R ²	66.3%	66.3%	66.2%	65.7%	65.9%	73.8%	75.5%	75.8%	54.5%	75.6%

We estimate linear regressions to examine the relationship between originator-sponsor affiliation and deal performance as of August 2009 for 474 Alt-A residential MBS deals completed between 2003 and 2007. Cumulative net loss rate (%) is the sum of all losses of principal suffered until August 2009 divided by the original pool balance. Foreclosure rate (%) is defined as the face value of all loans that are in the process of being foreclosed at the end of August 2009 as a percentage of the end of month pool balance. The definitions of other variables are in Appendix 2. We report coefficient estimates and t-statistics based on robust standard errors (in parentheses) in the table. We use (d) to denote that the variable is a dummy variable. ***, **, and * denote that the coefficient estimate is significantly different from zero at the 1%, 5%, and 10% level, respectively.

5.4 Table 4: Originator-sponsor affiliation and deal performance

Read:

- After controls, affiliated deals have **0.62 percentage points** lower cumulative net loss and **1.37 percentage points** lower foreclosure rates.
- The continuous share originated by the sponsor is also negatively related to losses and foreclosures.
- In low-doc deals, the affiliated coefficient is much larger: **-1.510** for cumulative net loss and **-2.83** for foreclosure rate.
- In high-doc deals, affiliation is statistically insignificant.

Economic message. The core result survives controls and appears exactly where the screening-incentive channel predicts: low-documentation deals.

5.5 Table 5: Propensity-score matching

Read:

- Panel A shows that affiliation is systematically related to observable deal characteristics.
- The authors match **191** affiliated deals to similar unaffiliated/mixed deals.
- In the matched sample, affiliation lowers cumulative net loss by **0.84 percentage points** in the full sample and **1.93 percentage points** in the low-doc sample.
- The matched treatment effect is insignificant for high-doc deals.

Economic message. The performance gap is not just an artifact of affiliated deals having better observable collateral.

5.6 Table 6: Instrumental-variables regressions

Read:

- The instrument is the sponsor's average affiliation rate in its previous sample deals.
- The instrument strongly predicts current affiliation.
- IV estimates imply that affiliation lowers cumulative net loss and foreclosure rates.
- The IV effect is concentrated in low-doc deals, while high-doc estimates are insignificant.

Economic message. The IV results suggest the OLS estimates are not simply upward biased by favorable unobservables. If anything, unobservable risk may make affiliated deals look riskier ex ante.

Table 5
Propensity score analysis

Panel A: Determinants of affiliation						
Explanatory variables	Pr(Affiliated=1)					
Low documentation (d)	0.24*** (-4.40)					
FICO	1.01 (0.69)					
LTV	0.80** (-2.40)					
% adjustable rate	0.98*** (-4.86)					
Negative amortization (d)	0.62 (-0.64)					
% purchase loans	0.94*** (-2.77)					
% family owned	0.92*** (-2.64)					
% primary residence	0.97** (-2.35)					
House price change	0.93** (-2.06)					
Vintage dummies	Yes					
Number of observations	474					
Pseudo R ²	38.1%					
Panel B: Treatment effects						
	Cumulative net loss rate (%)			Foreclosure rate (%)		
	All	Low-doc	High-doc	All	Low-doc	High-doc
Differences in loss rates, affiliated minus unaffiliated/mixed						
Unmatched sample, average	-3.05***	-3.36***	-1.26***	-6.68***	-5.11***	-3.70***
t-stat	(-11.50)	(-7.43)	(-4.33)	(-11.12)	(-6.03)	(-4.50)
Matched sample, ATT	-0.84*	-1.93***	-0.38	-2.59**	-5.52***	-1.12
t-stat	(-1.88)	(-2.83)	(-1.13)	(-2.05)	(-3.59)	(-0.85)

This table presents the results from a propensity score matching analysis where we match affiliated and unaffiliated/mixed MBS deals based on propensity scores calculated using logit regressions presented in Panel A. Logit results include odds ratios and robust *t*-statistics based on standard errors (*t*-statistics are negative when the odds ratio is less than one). We use the PSMATCH2 routine in STATA for matching. We use the nearest neighbor matching technique and match each affiliated deal with an unaffiliated/mixed deal based on their propensity scores. Also, we match with replacement and impose a common support dropping treatment observations whose propensity score is higher than the maximum or less than the minimum propensity score of the controls. We also limit the maximum distance between the propensity scores of the treatment and matched control firms to 1%. Average treatment effect of affiliation on affiliated deals (ATT) is presented in Panel B. ***, **, and * denote that the coefficient estimate in Panel A (or the estimated treatment effect in Panel B) is significantly different from zero at the 1%, 5%, and 10% level, respectively. Variable definitions are in Appendix 2.

Table 6
Instrumental variables (IV) regressions

Panel A: IV regressions for the overall sample					
Explanatory variables	First-stage Probit (1)	Second-stage			
		Cumulative loss rate (%) (2)	Foreclosure rate (%) (3)	Cumulative loss rate (%) (4)	Foreclosure rate (%) (5)
Average affiliation rate in past deals	4.128*** (5.51)	—	—	—	—
Affiliated (d)	—	-1.208*** (-3.59)	-2.394*** (-3.86)	-2.339*** (-6.34)	-3.657*** (-2.25)
Top 10 MBS sponsor (d)	—	—	—	(0.24) (0.55)	(0.98) (1.23)
Low documentation (d)	0.436** (2.38)	0.486** (2.35)	1.120** (2.49)	0.251** (1.26)	0.804** (1.68)
FICO	0.028** (2.53)	-0.044*** (-4.10)	-0.149*** (-7.33)	-0.032*** (-3.42)	-0.143*** (-2.17)
LTV	0.028 (0.35)	0.171*** (2.96)	0.148** (2.37)	0.078*** (1.87)	0.070** (1.04)
% adjustable rate	-0.005*** (-3.72)	0.016*** (5.03)	0.049*** (7.12)	0.016*** (5.41)	0.055*** (6.83)
Negative amortization (d)	-1.027*** (-5.86)	-1.240** (-2.40)	-3.724*** (-3.66)	-1.012** (-2.00)	-2.797*** (-2.45)
% purchase loans	-0.048*** (-7.34)	0.017 (1.08)	0.004 (0.13)	0.045 (3.07)	0.057 (1.78)
% family owned	-0.039** (-2.40)	0.021 (0.90)	0.022 (0.57)	0.026 (0.97)	0.024 (0.45)
% primary residence	-0.005 (-0.59)	-0.040*** (-5.08)	-0.025* (-1.70)	-0.065*** (-4.73)	-0.035* (-1.30)
House price change	0.036** (1.99)	-0.102*** (-4.34)	-0.238*** (-4.88)	-0.101*** (-4.28)	-0.222*** (-4.23)
Sponsor fixed effects	No	No	No	Yes	Yes
Number of observations	453	453	453	453	453

Panel B: IV regressions for the “low-doc” and “high-doc” subsamples				
Explanatory variables	Low-doc		High-doc	
	Cumulative loss rate (%) (1)	Foreclosure rate (%) (2)	Cumulative loss rate (%) (3)	Foreclosure rate (%) (4)
Affiliated (d)	-2.426*** (-6.70)	-4.811*** (-5.94)	0.238 (0.64)	0.011 (0.01)
Other controls	Yes	Yes	Yes	Yes
Number of observations	228	228	225	225

We estimate full maximum likelihood treatment-effects regressions and instrument the average affiliation rate in the sponsor’s past sample deals for affiliation to examine how originator-sponsor affiliation affects the ex post performance of our sample MBS deals. Since we need at least one past deal to calculate the instrument, we exclude the first deal of each sponsor from the estimation, which leaves us with an estimation sample of 453 deals. We present *t*-statistics based on robust standard errors in parentheses. We use (d) to denote that the variable is a dummy variable. ***, **, and * denote that the coefficient estimate is significantly different from zero at the 1%, 5%, and 10% level, respectively. Variable definitions are in Appendix 2.

5.7 Table 7: Loan-level default by affiliation and documentation

Table 7
Loan-level performance by affiliation and level of documentation

Panel A: Originator-sponsor affiliation and likelihood of default								
Explanatory variables	Pr(Default = 1)							
	All loans			Low documentation loans			Full documentation loans	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Originator affiliated with sponsor (d)	0.788*	0.894*	0.780*	0.766*	0.882*	0.763*	0.995	1.052
	(-13.05)	(-4.56)	(-13.80)	(-14.05)	(-4.86)	(-11.82)	(-0.15)	(-0.94)
Originator cannot sponsor (d)	—	0.977	—	—	0.891	—	—	0.923
	—	(-0.70)	—	—	(-0.27)	—	—	(-1.37)
Low documentation (d)	—	2.712*	—	—	—	—	—	—
	(9.36)	(7.39)	(9.36)	—	—	—	—	—
FICO	0.990*	0.990*	0.990*	0.990*	0.990*	0.990*	0.987*	0.987*
	(-55.76)	(-55.55)	(-55.86)	(-55.22)	(-55.86)	(-55.39)	(-30.73)	(-30.62)
LTV	1.054*	1.056*	1.054*	1.055*	1.057*	1.055*	1.044*	1.044*
	(26.22)	(27.09)	(26.24)	(25.11)	(26.07)	(25.11)	(28.43)	(27.33)
Adjustable rate mortgage (d)	1.501*	1.495*	1.499*	1.495*	1.480*	1.494*	1.507*	1.505*
	(17.64)	(18.01)	(17.54)	(16.73)	(14.61)	(16.68)	(13.08)	(11.08)
Negative amortization (d)	1.018	1.007	1.016	1.006	1.005*	1.005	1.069	1.222*
	(0.72)	(0.82)	(0.63)	(0.24)	(3.33)	(0.21)	(1.05)	(3.12)
Purchase loan (d)	0.759*	0.746*	0.759*	0.744*	0.739*	0.746*	0.737*	0.768*
	(-10.27)	(-10.32)	(-10.20)	(-10.58)	(-11.01)	(-10.96)	(-4.50)	(-4.42)
Family owned (d)	0.959	0.966	0.959	0.966	0.976	0.966	0.878	0.865*
	(-0.84)	(-0.71)	(-0.84)	(-0.84)	(-0.50)	(-0.68)	(-2.46)	(-2.46)
Owner occupied (d)	0.802*	0.803*	0.802*	0.809*	0.810*	0.809*	0.732*	0.721*
	(-5.02)	(-4.98)	(-5.00)	(-4.52)	(-5.00)	(-4.93)	(-4.60)	(-4.61)
MSA-level house price change	0.145*	0.138*	0.145*	0.145*	0.135*	0.145*	0.160*	0.162*
	(-8.24)	(-8.70)	(-8.23)	(-8.03)	(-8.54)	(-8.03)	(-8.50)	(-8.59)
Prepayment penalty (d)	1.305*	1.242*	1.307*	1.313*	1.255*	1.313*	1.345*	1.350*
	(10.62)	(7.79)	(10.35)	(11.21)	(8.42)	(10.89)	(5.90)	(3.12)
Silent second (d)	0.901*	0.885*	0.901*	0.910*	0.900*	0.910*	0.854*	0.850*
	(-6.08)	(-7.04)	(-6.05)	(-5.27)	(-6.61)	(-5.27)	(-4.41)	(-3.63)
Vintage fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Originator fixed effects	Yes	No	No	No	Yes	No	Yes	No
Number of observations	373,871	347,559	373,871	316,740	295,104	316,740	57,131	52,444
Pseudo R ²	15.3%	15.5%	15.3%	14.2%	14.5%	14.2%	12.9%	12.9%

(continued)

Table 7
Continued

Panel B: Propensity score matching			
Differences in ex post default rates, affiliated minus unaffiliated	All loans		
	Low documentation loans		Full documentation loans
Unmatched sample, average	-14.93*	-15.79*	-7.05*
t-stat	(-48.73)	(-43.41)	(-22.44)
Matched sample, average treatment effect of treated (ATT)	-5.09*	-4.79*	-5.09*
t-stat	(-13.58)	(-11.57)	(-1.72)

This table presents loan-level analysis of the relationship between originator-sponsor affiliation and performance. Because originator name is generally missing in ABSNet for loans securitized prior to 2006, we provide the results only for loans that were securitized in 2006 and 2007 and have non-missing originator name. Panel A presents logit estimates of the relation between originator-sponsor affiliation and likelihood of default (i.e., 90+ days delinquency from securitization until August 2009 for the overall sample and by level of documentation. We report odds ratios along with *t*-statistics based on robust standard errors clustered by MSA (i.e., statistics are negative when the odds ratio is less than one). In Panel B, we present average treatment effects for affiliation for affiliated loans, estimated based on a matching analysis. In particular, we match affiliated and unaffiliated loans based on propensity scores calculated using FICO score, cumulative loan-to-value ratio, adjustable rate dummy, negative amortization dummy, purchase loan dummy, family-owned dummy, primary residence dummy, prepayment penalty dummy, silent second dummy, one-year pre-deal house price run-up, vintage, and state in which the loan is originated. We estimate propensity scores using logit regressions. We use the PSMATCH2 routine in STATA for matching and calculating treatment effects. We use the nearest neighbor matching technique and matched affiliated loans with unaffiliated loans based on their propensity scores. We match with replacement. We impose a common support dropping treatment observations whose propensity score is higher than the maximum or less than the minimum propensity score of the controls. We also limit the maximum distance between the propensity scores of the treatment and control to 1%. We use (d) to denote that the variable is a dummy variable. * denotes that the coefficient estimate or the treatment effect is significantly different from zero at the 1% level. Variable definitions are in Appendix 2.

Read:

- The loan-level sample contains **364,709** affiliated loans and **115,387** unaffiliated loans.
- Raw loan default rates are **25.1%** for affiliated loans and **37.2%** for unaffiliated loans.
- For low-documentation loans, affiliation lowers default likelihood by about **23.4%** in the basic logit model.
- With originator fixed effects, affiliation is still associated with an **11.8%** lower likelihood of default.

- Loan-level propensity-score matching shows a **4.7 percentage point** lower default rate for affiliated low-doc loans, or about **16.8%** at the mean.

Economic message. The result is not driven by deal-level aggregation or differences across originators alone. Even within originators, affiliated low-doc loans perform better.

5.8 Table 8: Difference-in-differences around originator acquisitions

Read:

- The acquisitions studied are MortgageIT by Deutsche Bank and First Franklin by Merrill Lynch.
- The key interaction is *AcquiringSponsor* $\times$ *PostAcquisition*.
- For low-documentation loans, the interaction odds ratio is **0.700**, implying about a **30%** lower default likelihood after the loans become affiliated with the acquiring sponsor.
- There is no significant effect for full-documentation loans.

Economic message. When organizational affiliation changes, low-doc loan performance changes in the predicted direction. This is one of the paper’s strongest causal pieces of evidence.

Table 8
Difference-in-differences logit regressions

Explanatory variables	Pr(Default = 1)		
	All loans (1)	Low documentation loans (2)	Full documentation loans (3)
Acquiring sponsor (d)	1.387** (2.45)	1.363** (2.16)	1.184 (0.77)
Post-acquisition (d)	1.197** (2.46)	1.268*** (3.09)	0.864 (-0.89)
Acquiring sponsor (d) x Post-acquisition (d)	0.728* (-1.85)	0.700** (-2.00)	1.146 (0.43)
Low documentation (d)	3.064*** (17.39)	—	—
FICO	0.987*** (-22.30)	0.988*** (-20.01)	0.984*** (-13.69)
LTV	1.065*** (15.79)	1.065*** (15.84)	1.059*** (6.90)
Adjustable-rate mortgage (d)	1.422*** (5.47)	1.359*** (4.36)	1.801*** (4.10)
Negative amortization (d)	0.528*** (-7.80)	0.552*** (-7.05)	0.458*** (-2.98)
Purchase loan (d)	0.757*** (-4.33)	0.774*** (-3.76)	0.667*** (-3.30)
Family owned (d)	1.000 (0.01)	1.011 (0.15)	0.944 (-0.54)
Owner occupied (d)	0.718*** (-4.77)	0.722*** (-4.28)	0.705** (-2.37)
MSA-level house price change	0.089*** (-10.36)	0.090*** (-9.57)	0.088*** (-7.97)
Prepayment penalty (d)	1.391*** (4.19)	1.389*** (4.12)	1.356 (1.41)
Silent second (d)	0.754*** (-5.88)	0.740*** (-5.92)	0.908 (-0.74)
Number of observations	16,381	12,591	3,790
Pseudo R^2	17.2%	14.5%	19.1%

We estimate logit models to examine the determinants of default, defined as 60+ days delinquency, from securitization until August 2009. The sample is limited to Alt-A mortgages originated by MortgageIT and First Franklin and securitized in 2006 and 2007. *Acquiring Sponsor* is a dummy variable that takes on the value of one if the loan was sold to a deal sponsored by the acquiring entity before or after the acquisition and zero otherwise. *Post Acquisition* is a dummy variable that equals one if the loan was sold after the acquisition of the originator. Prior to the acquisition all loans are classified as unaffiliated, but we distinguish among these loans by whether or not they were sold to deals sponsored by the future acquirer. After the acquisitions, some loans are classified as affiliated (they are sold to deals sponsored by the acquirer) and some loans are classified as unaffiliated. We report odds ratios and *t*-statistics based on robust standard errors clustered by MSA (in parentheses). *t*-statistics are negative when the odds ratio is less than one. We use (d) to denote that the explanatory variable is a dummy variable. ***, **, and * denote that the estimated odds ratio is statistically significant at the 1%, 5%, and 10% level, respectively.

Table 9
Originator-servicer and servicer/sponsor affiliation and deal performance

Explanatory variables	Panel A: Cumulative net loss rate (%)					
	Originator-servicer affiliation			Sponsor-servicer affiliation		
	All (1)	Low-doc (2)	High-doc (3)	All (4)	Low-doc (5)	High-doc (6)
% serviced by the originator	-2.20*** (-4.16)	-3.82*** (-5.26)	-0.31 (-0.49)	—	—	—
% serviced by the sponsor	—	—	—	0.79 (1.28)	2.19** (2.31)	-0.60 (-0.82)
Low documentation (d)	2.30*** (4.41)	—	—	2.21*** (3.88)	—	—
FICO	0.01 (0.57)	-0.05 (-1.29)	-0.04 (-1.46)	-0.01 (-0.17)	-0.09 (-1.59)	-0.05* (-1.81)
LTV	0.26*** (2.61)	0.37* (1.79)	0.20** (2.40)	0.23** (2.17)	0.43 (1.56)	0.17** (2.03)
% adjustable rate	0.02** (2.17)	0.10*** (4.89)	0.02** (2.45)	0.02** (2.16)	0.10*** (3.48)	0.02*** (2.59)
Negative amortization (d)	-0.27 (-0.21)	0.23 (0.15)	-3.18** (-2.13)	-0.56 (-0.39)	0.16 (0.08)	-3.72** (-2.44)
% purchase loans	0.03 (0.76)	0.01 (0.13)	-0.04 (-0.73)	0.02 (0.33)	-0.00 (-0.03)	-0.06 (-1.11)
% family owned	0.05 (0.74)	-0.10 (-1.19)	0.03 (0.31)	-0.01 (-0.13)	-0.23** (-2.51)	-0.00 (-0.00)
% primary residence	-0.07*** (-2.93)	-0.06* (-1.80)	-0.06** (-2.11)	-0.07** (-2.48)	-0.09** (-2.01)	-0.05* (-1.75)
House price change	-0.13* (-1.90)	0.51*** (2.59)	-0.10* (-1.73)	-0.11 (-1.49)	0.37 (1.37)	-0.08 (-1.53)
Vintage dummies	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	96	48	48	96	48	48
Adjusted R^2	67.4%	74.7%	68.8%	61.3%	60.3%	69.2%

Explanatory variables	Panel B: Foreclosure rate (%)					
	All (1)	Low-doc (2)	High-doc (3)	All (4)	Low-doc (5)	High-doc (6)
% serviced by the originator	-1.88* (-1.67)	-3.99*** (-2.36)	-0.88 (-0.55)	—	—	—
% serviced by the sponsor	—	—	—	1.04 (0.72)	4.22 (1.58)	-0.73 (-0.39)
Vintage dummies and other controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	96	48	48	96	48	48
Adjusted R^2	76.2%	73.4%	79.9%	65.3%	39.4%	79.8%

We estimate ordinary least squares (OLS) regressions that examine the relation between originator-servicer and servicer-sponsor affiliation and deal performance as of August 2009 for 96 Alt-A residential MBS deals completed between 2003 and 2007 in which the originator and sponsor were unaffiliated. We present the results for the entire sample as well as “low-doc” and “high-doc” subsamples. We report coefficient estimates and *t*-statistics based on robust standard errors in the table. In Panel A (Panel B), cumulative net loss rate (foreclosure rate) is the dependent variable. For brevity, in Panel B, we only report the coefficient of the affiliation measures, although the models are estimated with all the control variables reported in Panel A. We use (d) to denote that the variable is a dummy variable. ***, **, and * denote that the coefficient estimate is significantly different from zero at the 1%, 5%, and 10% level, respectively. Variable definitions are in Appendix 2.

5.9 Table 9: Originator-servicer and sponsor-servicer affiliation

Read:

Table 10

Summary statistics for deal structure variables

Variables	All deals				Affiliated deals				Unaffiliated and mixed deals			
	N	Mean	Median	SD	N	Mean	Median	SD	N	Mean	Median	SD
Initial average yield spread (bps)	416	202.5	202.9	55.3	204	187.6	193.6	46.9	212	216.9	220.5	58.9
Subordination	474	6.24%	6.00%	2.85%	231	4.82%	4.50%	2.40%	243	7.59%	7.30%	2.54%
Overcollateralization (d)	474	35.4%	—	—	231	10.0%	—	—	243	59.7%	—	—
Overcollateralization target	168	0.70%	0.66%	0.40%	23	0.65%	0.50%	0.38%	145	0.71%	0.60%	0.41%

This table presents summary statistics for MBS yields and credit enhancement measures by originator-sponsor affiliation. Variable definitions are in Appendix 2.

- The analysis focuses on **96** deals where originator and sponsor are unaffiliated, so sponsor affiliation does not confound the servicing channel.
- A higher share serviced by the originator predicts lower cumulative net loss and lower foreclosure rates.
- For low-doc deals, a one-standard-deviation increase in the share serviced by the originator is associated with a large reduction in losses.
- Sponsor-servicer affiliation does not show the same beneficial pattern.

Economic message. Retained servicing rights appear to create another form of skin in the game for the originator, especially where screening matters.

5.10 Table 10: Summary statistics for deal pricing and structure

Read:

- The average initial yield spread is **187.6 bps** for affiliated deals and **216.9 bps** for unaffiliated/mixed deals.
- Subordination is **4.82%** for affiliated deals and **7.59%** for unaffiliated/mixed deals.
- Overcollateralization appears in only **10.0%** of affiliated deals but **59.7%** of unaffiliated/mixed deals.

Economic message. Ex ante market structure treats affiliated deals as safer: investors require lower yields and rating structures require less credit protection.

5.11 Table 11: Affiliation and MBS deal structure

Table 11

Affiliation and MBS deal structure

Variables	Panel A: Originator-sponsor affiliation and MBS structure											
	Initial average yield spread (%)			Subordination (%)			Pr(Overcollateralization=1)			Overcollateralization target		
	All	Low-doc	High-doc	All	Low-doc	High-doc	All	Low-doc	High-doc	All	Low-doc	High-doc
Explanatory variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Affiliated (d)	-0.128*	-0.120***	0.033	-0.328*	-0.451**	-0.106	0.238***	0.424**	0.010**	-0.153*	-0.221**	0.159
	(-1.81)	(-2.90)	(0.40)	(-1.68)	(-2.52)	(-0.25)	(-4.15)	(-2.08)	(-3.90)	(-1.83)	(-2.37)	(0.66)
Low documentation (d)	0.024	—	—	0.234	—	—	1.097	—	—	0.044	—	—
	(0.60)	—	—	(1.23)	—	—	(0.26)	—	—	(0.61)	—	—
FICO	-0.012**	-0.017***	-0.007***	-0.045***	-0.063***	-0.032**	0.964**	0.997	0.478***	0.002	-0.002	0.004
	(-2.57)	(-5.37)	(-2.58)	(-3.75)	(-4.80)	(-2.30)	(2.26)	(0.15)	(3.11)	(0.53)	(-0.30)	(0.57)
LTV	0.054***	0.027***	0.037***	0.260***	0.283***	0.249***	1.215**	1.271**	1.201	0.022**	0.027**	-0.010
	(4.46)	(2.85)	(4.07)	(9.95)	(7.26)	(2.34)	(2.36)	(2.51)	(0.80)	(2.47)	(2.43)	(-0.42)
% adjustable rate	-0.002***	-0.003***	-0.001	0.012***	0.009**	0.015***	1.006	1.006	1.015	-0.002**	-0.002*	0.001
	(-2.97)	(-3.15)	(-0.55)	(4.40)	(2.43)	(3.19)	(1.30)	(1.30)	(1.34)	(-2.00)	(-1.86)	(0.36)
Negative amortization (d)	0.181	0.223	1.109***	2.121***	1.444**	-0.033	8.659***	4.198	—	0.273*	0.389*	-0.028
	(1.32)	(1.19)	(4.46)	(9.71)	(2.14)	(-0.60)	(2.61)	(1.40)	—	(1.72)	(1.92)	(-0.10)
% purchase loans	0.004	0.004	0.004	0.008	-0.033	0.035**	1.045*	1.015	1.143**	0.002	0.002	-0.004
	(1.28)	(0.74)	(0.88)	(0.65)	(-1.49)	(2.39)	(1.82)	(0.49)	(1.99)	(0.50)	(0.38)	(-0.51)
% family owned	0.012***	-0.004	0.020**	0.040***	0.018	0.060***	0.984	0.930**	1.302***	-0.003	-0.011	0.004
	(2.84)	(-0.58)	(1.94)	(2.94)	(0.55)	(2.88)	(-0.48)	(-2.01)	(2.69)	(-0.59)	(-1.38)	(0.45)
% primary residence	-0.006***	-0.006***	-0.009	-0.027*	-0.002	0.088	1.097	0.920*	0.084	0.004	-0.010	-0.010
	(-2.80)	(-1.10)	(-3.11)	(-1.27)	(-1.95)	(-0.18)	(-0.75)	(-0.13)	(-2.06)	(1.10)	(0.96)	(-0.70)
Pre-deal house price run-up	-0.023**	-0.047***	0.005	0.061*	0.022	0.005	-0.044**	0.025	0.883	-0.022*	-0.025*	-0.001
	(-2.30)	(-3.24)	(0.30)	(1.74)	(1.26)	(0.11)	(-2.00)	(-1.13)	(0.81)	(-1.70)	(-1.80)	(-0.02)
Vintage dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Model	OLS	OLS	OLS	OLS	OLS	OLS	Logit	Logit	Logit	OLS	OLS	OLS
Number of observations	416	209	207	474	237	237	474	231	254	168	128	40
Adjusted or Pseudo R ²	26.3%	24.4%	25.7%	64.6%	61.4%	55.7%	44.9%	24.4%	68.5%	26.9%	26.9%	36.9%

(continued)

Table 11

Continued

Variables	Panel B: Originator-servicer affiliation and MBS structure using deals in which the originator and sponsor are not affiliated											
	Initial average yield spread (%)			Subordination (%)			Pr(Overcollateralization=1)			Overcollateralization target		
	All	Low-doc	High-doc	All	Low-doc	High-doc	All	Low-doc	High-doc	All	Low-doc	High-doc
Explanatory variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
% serviced by the originator	-0.258*	-0.457***	-0.024	-1.043**	-1.283***	-0.975	2.556	1.036	0.079	0.035	0.024	0.120
	(-1.81)	(-2.78)	(-0.10)	(-2.06)	(-2.60)	(-0.90)	(1.03)	(1.09)	(-1.20)	(0.28)	(0.13)	(0.51)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Model	OLS	OLS	OLS	OLS	OLS	OLS	Logit	Logit	Logit	OLS	OLS	OLS
Number of observations	83	44	39	96	48	48	96	48	48	58	35	23
Adjusted or Pseudo R ²	38.6%	39.2%	42.1%	31.8%	67.5%	39.5%	42.7%	73.4%	54.6%	6.7%	12.1%	48.5%

We estimate how initial average yield spread, subordination, likelihood of overcollateralization, and overcollateralization target of the 474 sample MBS deals is related to originator-sponsor affiliation (Panel A) and originator-servicer affiliation (Panel B). We estimate likelihood of overcollateralization with logit (odds ratios) are reported in the table, the other models with OLS. Regressions in Panel A are based on the overall sample of 474 deals, and those in Panel B are based only on deals in which the originator(s) are unaffiliated with the sponsor. We report *t*-statistics based on robust standard errors in parentheses. In logit regressions, *t*-statistics are negative when the odds ratio is less than one. We use ***, **, and * to denote that the coefficient estimate is significantly different from zero at the 1%, 5%, and 10% level, respectively. Variable definitions are in Appendix 2.

Read:

Table 12
Yield spreads of AAA-rated fixed rate Alt-A mortgage-backed bonds at origination

Panel A: Summary statistics						
Deal type	Low-doc			High-doc		
	N	Mean	SD	N	Mean	SD
Affiliated (d) = 1	381	129.5	22.1	990	132.7	33.17
Affiliated (d) = 0	143	137.3	33.3	259	126.1	35.19
Difference		-7.8	-11.2		6.6	-2.02
<i>p</i> -value from <i>t</i> -test for equal means		0.00			0.01	
<i>p</i> -value from <i>F</i> -test for equal variances			0.00			0.22

Panel B: Regression results		
Explanatory variables	Yield spread (in basis points)	
	Low-doc (1)	High-doc (2)
Affiliated (d)	-11.16*** (-3.56)	2.10 (1.04)
Weighted average life (WAL)	-0.57** (-2.52)	-2.49*** (-10.68)
Weighted average coupon rate (WAC)	32.33*** (7.10)	31.83*** (10.22)
Vintage fixed effects	Yes	Yes
Bond type fixed effects	Yes	Yes
Number of observations	524	1,249
Adjusted R^2	18.5%	32.1%

We examine the relationship between originator sponsor affiliation and initial yield spreads of AAA-rated mortgage-backed bonds. The sample is restricted to fixed rate bonds. We obtained the coupon rates of bonds as well as estimated (by the sponsor) number of years it will take to pay half of the bond principal (WAL) and average coupon rate of the underlying collateral mortgages weighted by mortgage balance (WAC) from Bloomberg. Initial yield spread is defined as the difference between the initial fixed coupon rate of the bond and the yield on the constant maturity Treasury bond that has the closest maturity to the WAL of the bond. We set the top and bottom 5% of the spread to missing because of obvious data errors (e.g., negative initial yield spread). Variable definitions are in Appendix 2.

- In low-doc deals, originator–sponsor affiliation lowers initial average yield spreads by about **32 bps**, or **14.8%** of the mean.
- Low-doc affiliated deals require **0.65 percentage points** less subordination, about **8.7%** of the mean subordination rate.
- Affiliated low-doc deals are much less likely to use overcollateralization and have smaller OC targets when an OC account exists.
- Originator-servicer affiliation also predicts lower yields and lower subordination in low-doc un-affiliated deals.

Economic message. The same incentive variables that predict lower realized losses also predict cheaper and less protected MBS structures at issuance.

5.12 Table 12: Yield spreads of fixed-rate AAA bonds

Read:

- The paper studies **1,773** fixed-rate AAA bonds.
- In low-doc deals, affiliated bonds have lower raw spreads: **129.5 bps** versus **137.3 bps**.
- In regressions with weighted average life, weighted average coupon, vintage fixed effects, and bond-type fixed effects, affiliated low-doc AAA bonds have spreads **11.16 bps** lower.
- In high-doc deals, the affiliation coefficient is insignificant.

Economic message. Pricing differences are not only due to affiliated deals issuing more AAA securities. Even within AAA-rated fixed-rate tranches, investors demand lower yields when low-doc deals have originator skin in the game.

6 Short summary

- The paper studies whether originator retained exposure matters in mortgage securitization.
- The main proxy for retained exposure is originator affiliation with the MBS sponsor or servicer.
- Affiliated Alt-A deals have lower cumulative losses and foreclosure rates.
- The performance advantage is concentrated in low-documentation loans and deals.
- Propensity-score matching, IV regressions, loan-level controls, originator fixed effects, and acquisition-based difference-in-differences all support the same conclusion.
- Investors and rating agencies appear to recognize these incentives ex ante: affiliated low-doc deals have lower yields and need less credit enhancement.
- The main implication is that securitization incentives depend on organizational form and retained exposure, not only on whether loans are securitized.

7 Conclusion

7.1 Core conclusion

Demiroglu and James show that “skin in the game” is economically important inside the mortgage-backed-securities market. When the originator is affiliated with the sponsor or retains servicing rights, low-documentation loans perform better ex post.

7.2 Economic intuition

Low-documentation loans make screening more valuable because hard borrower information is limited. If the originator expects to retain some downside exposure through sponsor affiliation or servicing rights, the originator has stronger incentives to screen borrowers carefully. That better screening shows up later as lower defaults, losses, and foreclosures.

7.3 Takeaway

The paper’s message is that incentive alignment mattered and was at least partly recognized by the market. Investors and rating agencies did not treat all securitized loans as identical: they demanded higher yields and more credit enhancement when the originator had less skin in the game.